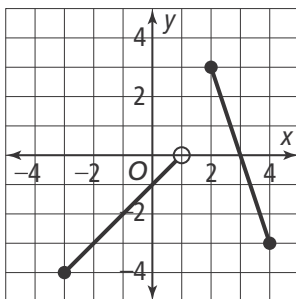


1-3 Lesson Quiz

Piecewise-Defined Functions

1. Which rule defines the function in the graph?



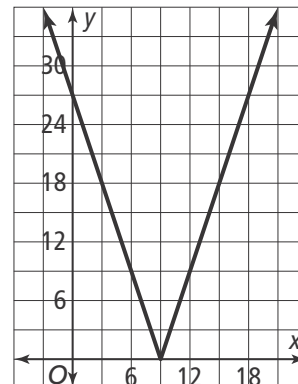
- Ⓐ $f(x) = \begin{cases} x - 1, & \text{if } -3 < x \leq 1 \\ -3x + 9, & \text{if } 2 < x < 4 \end{cases}$
- Ⓑ $f(x) = \begin{cases} x - 1, & \text{if } -3 \leq x < 1 \\ -3x + 9, & \text{if } 2 \leq x \leq 4 \end{cases}$
- Ⓒ $f(x) = \begin{cases} x - 1, & \text{if } -3 \leq x < 1 \\ x - 3, & \text{if } 2 \leq x \leq 4 \end{cases}$
- Ⓓ $f(x) = \begin{cases} x - 1, & \text{if } -3 < x \leq 1 \\ x - 3, & \text{if } 2 < x < 4 \end{cases}$

2. A scuba diver returning to the surface swims to move up quickly, then waits a few meters below the surface to avoid pressure issues. They complete the ascent more slowly. The function f represents the diver's height above sea level over time, in seconds. Which of the following is true for f ?

$$f(x) = \begin{cases} 3x + 1, & -9 < x \leq -2 \\ -5, & -2 < x \leq 1 \\ x - 6, & 1 < x < 7 \end{cases}$$

- Ⓐ The domain is the set of times $-9 < x < -2$, $-2 < x < 1$, and $1 < x < 7$.
- Ⓑ The domain is the depth interval $(-26, 1)$.
- Ⓒ The domain is the time interval $(-9, 7)$.
- Ⓓ The domain is the depth interval $(-9, 1]$.

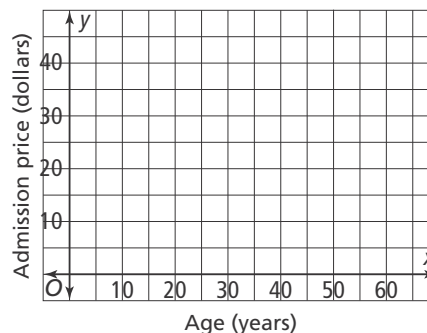
3. Write the piecewise-defined function shown in the graph.



$$f(x) = \begin{cases} \boxed{}(x + \boxed{}), & x \leq \boxed{} \\ \boxed{}(x + \boxed{}), & x > \boxed{} \end{cases}$$

4. The table shows the admission prices for an amusement park. Graph the function.

Age	Admission Price
under 5	free
5–17	\$25
18–64	\$40
65 and over	\$35



5. Identify the domain and range of the function in Item 4. Write the range values in ascending order.

domain: $x > \boxed{}$

range: $\{\boxed{}, \boxed{}, \boxed{}, \boxed{}\}$